FIBRES OVER A DECENT BASE POINT WITH BOUNDED DEGREES ARE FINITE UNIONS OF POINTS WITH FINITE RESIDUE EXTENSIONS

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Editorial note. The two uncarded chapter inputs are declared external.

Allowed standard prerequisites (editorial).

- Example schemes are decent. example-schemes (same chapter, NOT carded, declared external). The scheme case feeds the decent identification.
- Weak decency of points over fields. lemma-weakly-decent-over-field (same chapter, NOT carded, declared external). Weak decency of the fibre over a field.

Source reference. Original equation 0AC2: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4292–4292. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0AC1: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4282–4282. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AC3: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4298–4298. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AC5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4326–4326. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABM: focused-seeds/geometry/S9-decent-decent-finite-fibre/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03MI: focused-seeds/geometry/S9-decent-decent-finite-fibre/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In diagram (0AC1) the set (0AC2) is finite if y can be represented by a monomorphism $\mathrm{Spec}(k) \to Y$ where k is a field and g is quasi-finite at z. (Special case: Y is decent and g is étale.)*

Proof. By Lemma 0AC5 applied twice we may replace Z by $Z_k = \mathrm{Spec}(k) \times_Y Z$ and X by $X_k = \mathrm{Spec}(k) \times_Y X$. We may and do replace Y by $\mathrm{Spec}(k)$ as well. Note that $Z_k \to \mathrm{Spec}(k)$ is quasi-finite at z by Morphisms of Spaces, Lemma 0ABM. Choose a scheme V, a point $v \in V$, and an étale morphism $V \to Z_k$ mapping v to z. Choose a scheme U, a point $u \in U$, and an étale morphism $U \to X_k$ mapping u to x. Again by Lemma 0AC5 it suffices to show $F_{u,v}$ is finite for the diagram

$$\begin{array}{ccc} U \times_{\mathrm{Spec}(k)} V & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & \mathrm{Spec}(k) \end{array}$$

The morphism $V \to \mathrm{Spec}(k)$ is quasi-finite at v (follows from the general discussion in Morphisms of Spaces, Section 03MI and the definition of being quasi-finite at a point). At this point the finiteness follows from Example 0AC3. The parenthetical remark of the statement of the lemma follows from the fact that on decent spaces points are represented by monomorphisms from fields and from the fact that an étale morphism of algebraic spaces is locally quasi-finite. □